ADITYA THAKARE

Data Scientist | GenAI & Agentic AI | MLOps

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SUMMARY

Data Scientist with 3+ years of experience building and deploying Generative AI systems — including RAG pipelines, multi-agent LangGraph orchestration, and end-to-end MLOps workflows. Proficient in LangGraph, LangChain, FastAPI, Docker, and MLflow. Built and deployed four AI systems (legal AI, agentic research, predictive maintenance, financial analytics), including a production agentic analytics system built on contract for Neon AI. Delivered technical training to 2,000+ students and faculty across 60+ workshops and faculty development programs at 30+ institutions. Research background in deep learning-based ancient script recognition and applied NLP/ML.

SKILLS

Languages	Python, SQL
Generative AI	RAG · Agentic AI · LLMs · LangGraph · LangChain · Gemini API · OpenAI API · Prompt Engineering
Vector Stores & Embeddings	ChromaDB · Pinecone · Hugging Face · Gemini Embeddings · Semantic Chunking
ML / Deep Learning	Scikit-learn · XGBoost · PyTorch · TensorFlow · Keras · CNNs · NLP · Transfer Learning
Backend & APIs	FastAPI · REST APIs · Streamlit · Pydantic
MLOps & DevOps	Docker · Nginx · GitHub Actions · CI/CD · MLflow · DVC · systemd · SSL/HTTPS
Data & Visualization	Pandas · NumPy · Matplotlib · Seaborn · Power BI · Tableau

PROFESSIONAL EXPERIENCE

AI Consultant & Developer (Contract) | Neon AI

May 2026 – Jul 2026

Architected the agentic AI backend of a **production** financial analytics dashboard that answers C-suite executives' natural-language questions: **LangGraph** orchestration behind an **async FastAPI** serving layer.

Integrated **DuckDB/Parquet** for high-performance analytical queries and **MongoDB** for metadata; implemented **Pydantic** structured outputs with GPT-4o-mini for schema-validated responses to financial queries.

Tech: LangGraph · FastAPI · DuckDB · Parquet · MongoDB · Pydantic · GPT-4o-mini

Data Scientist | MITU Skillologies

Sep 2023 – Present | Pune, India

Designed and built **end-to-end AI systems** demonstrating LangChain RAG pipelines, neural network architectures, and MLOps workflows.

Assisted in deep learning research on **Brahmi script recognition** — conducted literature review across OCR and CNN methodologies; supported baseline implementation across four architectures (Custom Deep CNN, InceptionNet, ResNet-50, XceptionNet) on a **7,880-image, 56-class dataset**; ResNet-50 selected as optimal with **highest benchmark accuracy** across all evaluated models.

Contributed research and implementation support to applied ML projects including **student progression prediction** (multi-source data integration, feature analysis) and **crop disease classification** — comparative experimentation across multiple model architectures.

Primary technical SME across **60+ workshops and faculty development programs at 30+ institutions**, reaching **2,000+ students and faculty**; architected a **140-session full-stack AI curriculum** (Python through Generative AI and LangGraph) and led a 3-hour session on "Next-Gen AI: Neural Networks & Applications" for mixed technical and non-technical audiences.

OPEN SOURCE

caura-long-run-fleet | **caura-ai** — Merged PR #43 | Oct 2026

Added a configurable **--agent-timeout** CLI flag to the fleet simulator of an open-source orchestration and memory platform for long-running AI agents, wired through to the agent subprocess calls and documented in the README (fixes issue #36).

PROJECTS

Agentic Research System (Multi-Agent AI)

Live Demo | Source Code

Solves shallow, single-pass AI research outputs via a self-refining multi-agent system that iterates until quality thresholds are met.

Built and deployed a multi-agent system using **LangGraph** — chosen over LangChain's AgentExecutor for native cyclic graph support, persistent typed state, and deterministic control flow. Three specialized agents: Researcher (Tavily + Gemini synthesis), Writer (structured drafting), and Critic (quality scoring with revision trigger). **TypedDict state schema** with conditional edges loops back through Writer only when Critic score falls below threshold — preventing infinite loops and premature termination.

Tech: Agentic AI · LangGraph · LangChain · Gemini API · Tavily Search · Streamlit · Docker

Typed Decision Models: Agent Router Benchmark

Source Code

Benchmarks a typed decision model (Laya, Hugging Face) as the router step of an agent workflow.

Compared fine-tuned **Laya** against Random Forest, **XGBoost**, a fine-tuned **BERT** and a **Gemini** LLM router behind one DecisionModel interface with shared metrics and p50/p95 latency, inside a **LangGraph** workflow. On a synthetic benchmark, fine-tuned Laya beat the Gemini router by **25 points on exact match** (0.76 vs 0.51) at **~21x lower latency** and **~80x lower cost**, while XGBoost matched Laya at ~300x smaller size.

Tech: Hugging Face · LangGraph · BERT · XGBoost · Gemini API · Python · pytest

Conversational Legal Agent (RAG)

Live Demo | Source Code

Addresses the lack of affordable legal information access for individuals who cannot afford legal consultation.

Deployed a conversational RAG agent for Indian law that answers with cited sources, over **~70K chunks from 1,000+ Indian acts** (including the 2023 criminal codes). **Hybrid retrieval**: BM25 plus vector search merged with custom **Reciprocal Rank Fusion**, since legal queries mix exact section numbers with plain-language meaning; category-filtered **ChromaDB**, a classifier and a **reranker**. First self-hosted on DigitalOcean (Nginx, SSL, systemd), now on Streamlit Cloud.

Tech: RAG · Hybrid Search (BM25 + RRF) · ChromaDB · Gemini API · LangChain · Streamlit · Nginx

Industrial Equipment Maintenance (End-to-End MLOps)

Live Demo | Source Code

Targets the high cost of unplanned manufacturing downtime via sensor-driven predictive maintenance.

Developed a predictive maintenance classifier using **XGBoost** — selected over neural alternatives for tabular sensor data in low-data regimes. Built a reproducible **DVC** pipeline (ingest → featurize → train → evaluate) with **MLflow** experiment tracking. Fully decoupled architecture: **FastAPI** inference API deployed via **Docker** on **Render**, **Streamlit** dashboard on Streamlit Community Cloud, and **GitHub Actions** CI/CD (ruff lint + pytest on every push). Achieved **98.3% accuracy** with **91.2% F1** on imbalanced sensor data using scale_pos_weight tuning.

Tech: XGBoost · FastAPI · MLflow · DVC · Streamlit · Docker · GitHub Actions · Render

CERTIFICATIONS

Generative AI and Large Language Models — IIM Bangalore (SWAYAM) *Jul – Nov 2024*

Generative AI, LLM & RAG — GeeksforGeeks *Dec 2025*

Supervised Machine Learning: Regression and Classification — DeepLearning.AI, Stanford University *Nov 2024*

Prompt Engineering for ChatGPT — Vanderbilt University *Mar 2024*

Python Skill Up — GeeksforGeeks *Dec 2025*

Databases and SQL for Data Science with Python — Coursera *May 2023*

Docker Training Course for the Absolute Beginner — KodeKloud *Mar 2023*

EDUCATION

PG Diploma in Artificial Intelligence — CDAC ACTS, Pune *2023*

B.Tech – Civil Engineering (CGPA: 7.78) — Savitribai Phule Pune University *2021*